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Economic Prosperity and Happiness:
An Econometric Analysis Using the World Happiness Report (2015–2024)

I. Introduction

Economic well-being has traditionally been measured using aggregate indicators such as gross domestic product, income levels, and economic growth rates. While these measures capture important aspects of economic performance, they do not directly reflect how individuals perceive their quality of life. As a result, economists have increasingly turned toward subjective measures of well-being, including happiness and life satisfaction, to obtain a more comprehensive picture of societal welfare.

The emerging field of happiness economics challenges the assumption that higher income automatically translates into higher well-being. Although income allows individuals to meet basic needs and enjoy material comforts, numerous studies suggest that its marginal benefits diminish after a certain point. At the same time, non-economic factors such as health, social relationships, and institutional trust may play critical roles in shaping overall happiness.

This study contributes to this literature by empirically examining the determinants of national happiness using cross-country data from the World Happiness Report. Our central research question is: How does economic prosperity, measured by GDP per

capita, influence national happiness, and what role do health and social support play when income is held constant? Answering this question is particularly relevant for policymakers seeking to improve well-being beyond traditional economic objectives.

II. Research Question and Hypotheses

Our research question examines the relationship between economic, health, and social factors and subjective well-being at the national level. Specifically, we analyze whether income alone is sufficient to explain differences in happiness across countries or whether additional factors are equally important.

Based on economic theory and prior empirical research, we propose three hypotheses.

First, GDP per capita is expected to have a positive effect on happiness, as greater income allows individuals to satisfy basic needs and reduce economic insecurity. Second, healthy life expectancy is expected to increase happiness by improving longevity and quality of life. Third, social support is expected to have a strong positive effect on happiness, reflecting the importance of social networks, trust, and community support systems.

III. Data Description

The data used in this study are drawn from the World Happiness Report, covering the period from 2015 to 2024. The dataset compiles information from large-scale international surveys, along with economic and health indicators collected by international organizations such as the World Bank and the World Health Organization.

The unit of observation is a country-year, meaning each row corresponds to a specific country in a particular year. The final sample contains 1,502 observations and includes countries from all major geographic regions, providing substantial cross-country variation.

The dependent variable is the Happiness Score, which measures average self-reported life satisfaction on a standardized scale. The key independent variables include GDP per capita, Healthy Life Expectancy, and Social Support. GDP per capita serves as a proxy for economic prosperity, healthy life expectancy measures population health outcomes, and social support captures the availability of social networks and emotional assistance.

IV. Data Cleaning and Preparation

Before conducting the econometric analysis, several data cleaning and preparation steps

were required. The raw dataset used semicolons as delimiters rather than commas and employed European-style decimal formatting, where commas are used instead of periods. These issues were corrected to ensure proper parsing of the data.

All variables relevant to the analysis were converted to numerical data types. Observations containing missing values in any of the regression variables were removed to maintain a consistent sample across all estimations. After cleaning, the dataset contained 1,502 complete country-year observations.

In addition, variables were examined for extreme or implausible values. No observations were excluded based on outliers, as all values fell within reasonable ranges consistent with documentation from the World Happiness Report.

V. Descriptive Statistics

Descriptive statistics provide an overview of the distribution and variation of the variables used in the analysis. Table 1 reports the mean, standard deviation, minimum, and maximum values for each variable included in the regression model. These statistics highlight substantial variation in happiness, income, health, and social support across

countries.

VI. Exploratory Data Analysis

Exploratory data analysis was conducted to visually examine the distribution of national happiness and its relationship with key explanatory variables prior to formal econometric estimation. This step helps identify underlying patterns in the data and assess whether a multivariate regression framework is appropriate.

Figure 1 presents a histogram of happiness scores across all country-year observations. The distribution is approximately bell-shaped, with most countries clustered around moderate happiness levels. Only a small number of observations lie at very low or very high happiness levels, suggesting substantial variation in well-being across countries without extreme skewness. This supports the use of linear regression techniques for the analysis.

Figure 2 displays a scatter plot of GDP per capita against happiness score. The relationship is clearly positive, showing countries with higher income levels tend to report higher happiness. However, the slope of the relationship flattens at higher income levels, indicating diminishing marginal returns to income. While increases in income are strongly associated with gains in happiness among low-income countries, additional income in already wealthy countries is associated with smaller improvements in well-being.

Figure 3 presents a scatter plot of social support against happiness score. This figure shows a strong and approximately linear positive relationship. Countries with higher levels of social support consistently report higher happiness scores. Compared to GDP per capita, the relationship between social support and happiness appears tighter and more uniform across observations, suggesting that social connections may play a more robust role in explaining well-being than income alone.

Taken together, these visualizations indicate that while economic prosperity is positively associated with happiness, non-economic factors—particularly social support—play a

critical role in shaping well-being. The patterns observed in Figures 2 and 3 motivate the inclusion of health and social variables alongside income in the multivariate regression framework used in the subsequent analysis.



Figure 1: Histogram of Happiness Scores



Figure 2: GDP per Capita vs. Happiness Score

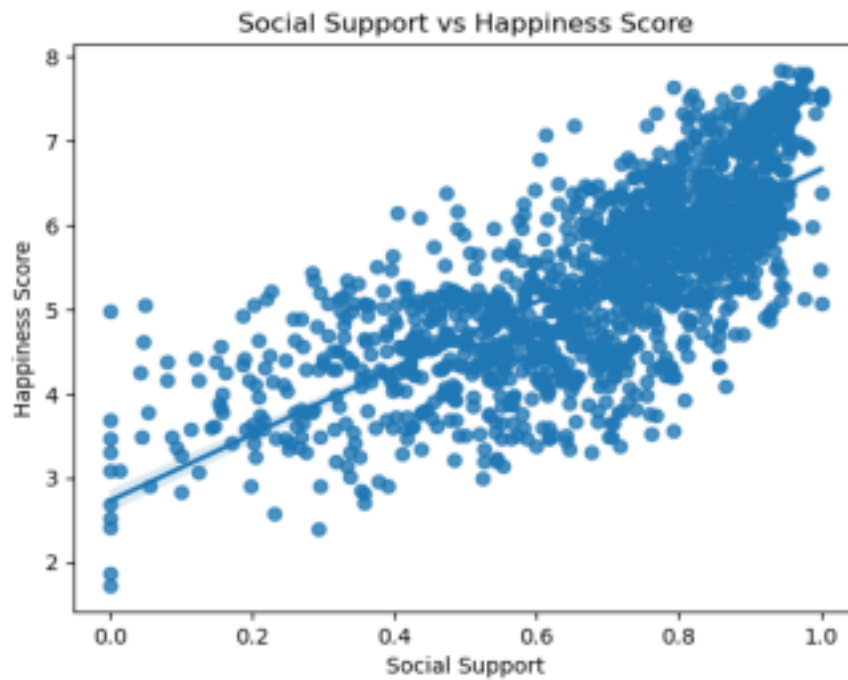


Figure 3: Social Support vs. Happiness Score

VII. Econometric Model and Results

To formally examine the determinants of national happiness, we estimate the following ordinary least squares (OLS) regression model:

$$\text{HappinessScore}_i = \beta_0 + \beta_1 \text{GDPperCapita}_i + \beta_2 \text{HealthyLifeExpectancy}_i + \beta_3 \text{SocialSupport}_i + u_i$$

This specification allows us to isolate the partial effect of economic prosperity on happiness while controlling for health and social factors. Table 2 presents the estimated regression coefficients along with standard errors, t-statistics, and p-values. The regression results indicate that all explanatory variables have positive and statistically significant effects on happiness. The coefficient on GDP per capita is positive but relatively modest in magnitude, supporting the idea of diminishing marginal utility of income. Healthy life expectancy has a meaningful positive effect, highlighting the importance of health outcomes. Social support has the largest coefficient in the model, underscoring the central role of social relationships in determining well-being.

The model achieves an R-squared of approximately 0.66, indicating that about 66 percent of the variation in happiness across countries and years is explained by the included variables. This represents a strong fit for cross-country data.

VIII. Discussion of Findings

The results of the econometric analysis highlight the multifaceted nature of happiness. While economic prosperity contributes positively to well-being, its effect is considerably smaller than that of social support and health. This finding suggests that policies aimed

solely at increasing income may not yield substantial improvements in happiness unless accompanied by investments in health systems and social infrastructure.

The strong effect of social support emphasizes the importance of social cohesion, trust, and interpersonal relationships. These factors may enhance resilience, reduce stress, and improve overall life satisfaction, even in the absence of high income levels.

IX. Limitations and Future Research

This study is subject to several limitations. First, the analysis may suffer from omitted variable bias, as factors such as cultural norms, institutional quality, and income inequality are not explicitly included. Second, the pooled cross-country design limits our ability to make causal claims.

Future research could address these limitations by incorporating additional control variables, applying panel data techniques such as fixed effects models, or exploring nonlinear relationships between income and happiness. Further work could also examine regional differences or heterogeneous effects across income groups.